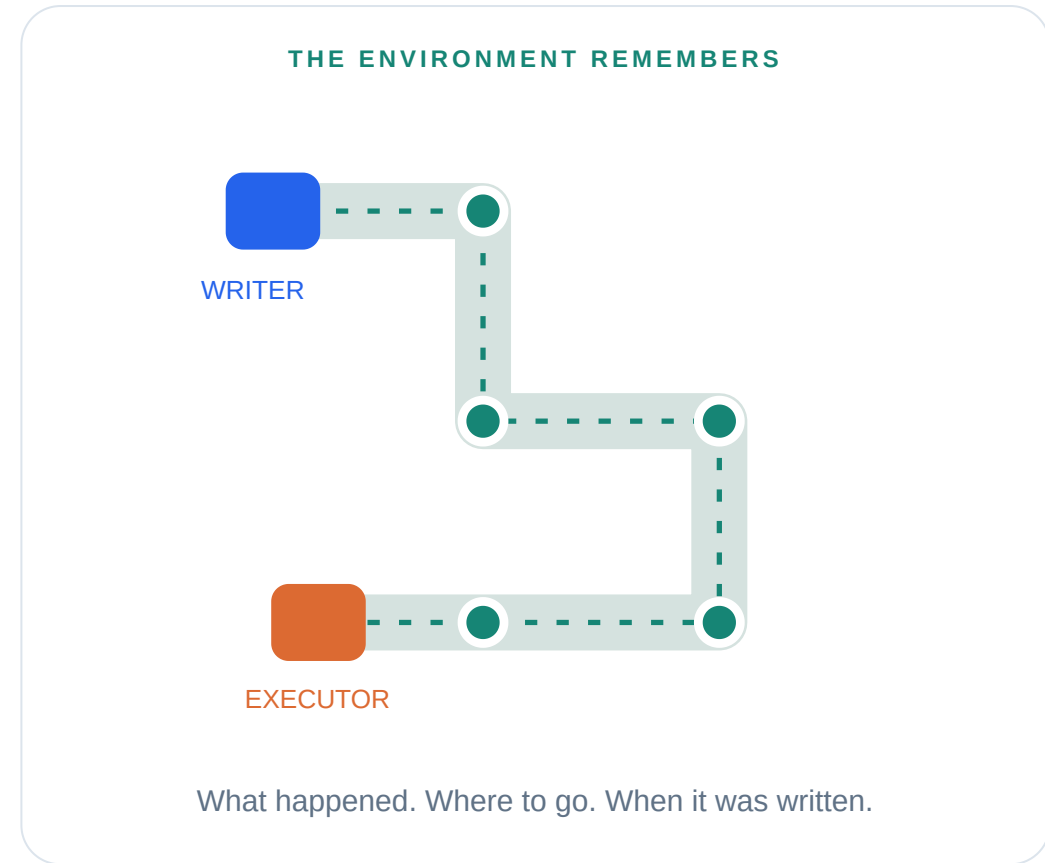


LivingMap.

If the Writer is lost, the map is not.

Spatial memory for emergency robots.

MINES & TUNNELS · PHASE 1 SIMULATION
LivingMap / Spatial memory for emergency robots



A robot can fail. Its discoveries should survive.

EXPLORING ROBOT

WRITER LOST

The exploring robot is no longer available.

System concept / implemented in the Writer-loss simulation scenario

INDEPENDENT MEMORY

B01

B02

B03

Beacons remain.

Event + guidance + timestamp
Persistent records. Relayed
advertisements.

MISSION CONTINUES

Executor inherits.

A gateway brief gives it memories and
the last uploaded map before entry.

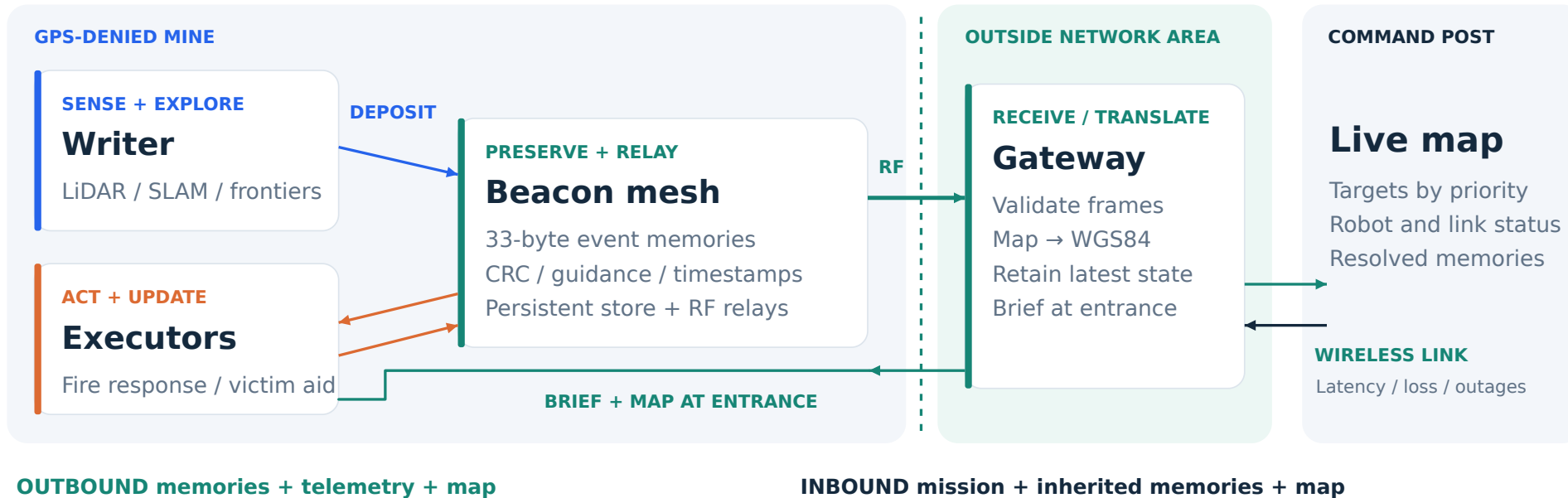
No restart from zero.

One boundary. One path to the outside.

01 / SYSTEM ARCHITECTURE

The mine keeps the memory.

One communication boundary. Three robot roles. Knowledge survives the Writer.



Telemetry, memories and map pass through the RF mesh / mission briefs return through the gateway. Command post: **WHO + WHAT**. Robot: **HOW**.

33 bytes. Three essential questions.

WHAT	WHERE	WHEN
Event, severity, confidence	Position, guidance, chain	Timestamp and expiry

03 / BEACON MESSAGE

33 bytes that answer three questions.

Version 2 · network byte order · CRC32 over the first 29 bytes

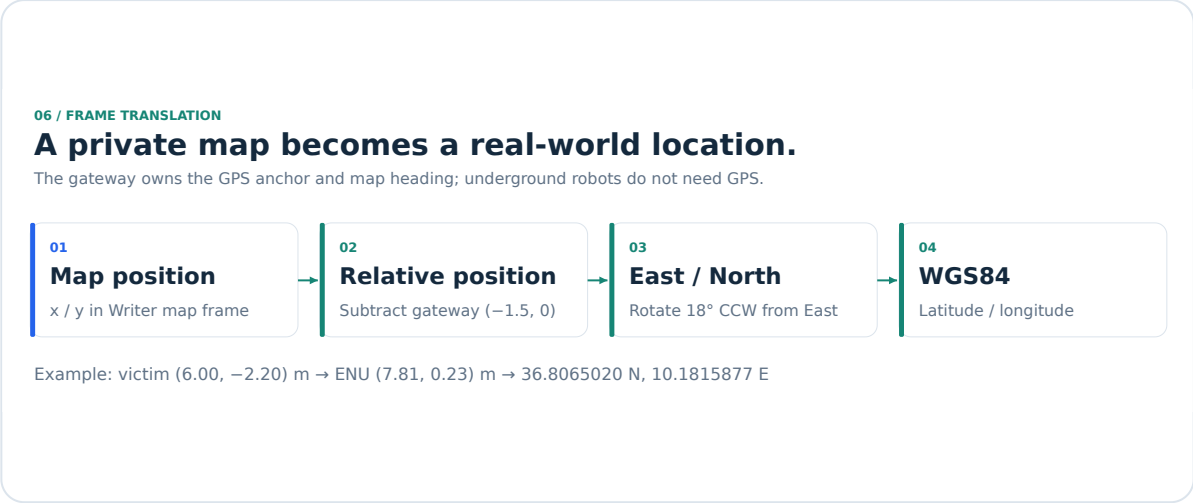
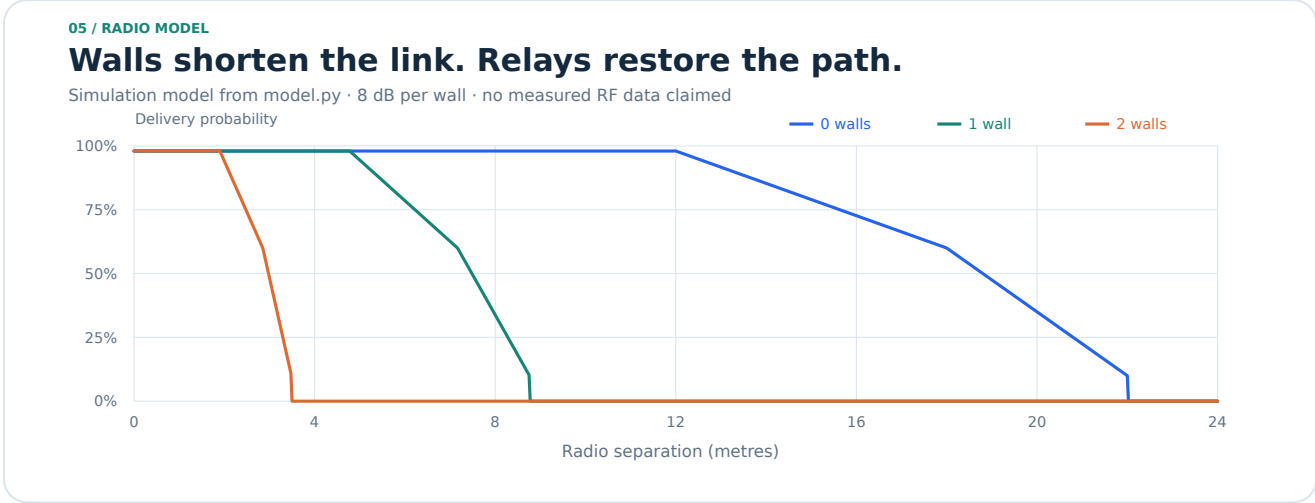
IDENTITY	WHEN	WHERE	WHERE	WHAT	WHERE	WHEN	INTEGRITY
8 B	4 B	4 B	4 B	3 B	4 B	2 B	4 B
Version / flags Mission / beacon / seq.	Unix timestamp uint32 seconds	Beacon x / y int16 centimetres	Bearing / range centi-deg. / cm	Type / severity Confidence %	Previous / next uint16 beacon IDs	Time to live uint16 seconds	CRC32 4-byte checksum
00-07	08-11	12-15	16-19	20-22	23-26	27-28	29-32

PROTECTED PAYLOAD / BYTES 00-28

Safe beacon position + guidance vector = event position. Previous leads out; next leads deeper in.

30-beacon capacity / 4 reserved for spares - CRC32 integrity over event tag.
A safe-standoff beacon records a reached place and points toward the event.

Relay through rock. Translate at the gateway.



RF mesh

8 dB modeled loss per wall.

Recorded memories arrive in 2–3 hops.

GPS outside

The gateway owns the GPS anchor.

Underground robots keep local coordinates.

Secure the scene. Then assist the victims.

07 / RECORDED MISSION

One inherited map. Three distinct roles.

5 October 2026 · spacious layout / one gas hazard · final snapshot, retained trails only



SOURCE docs/results/run_spacious_state.json · all panels use recorded map coordinates (metres)

WRITER

Explore + preserve

FIRE ROBOT

Extinguish + update

RESCUE

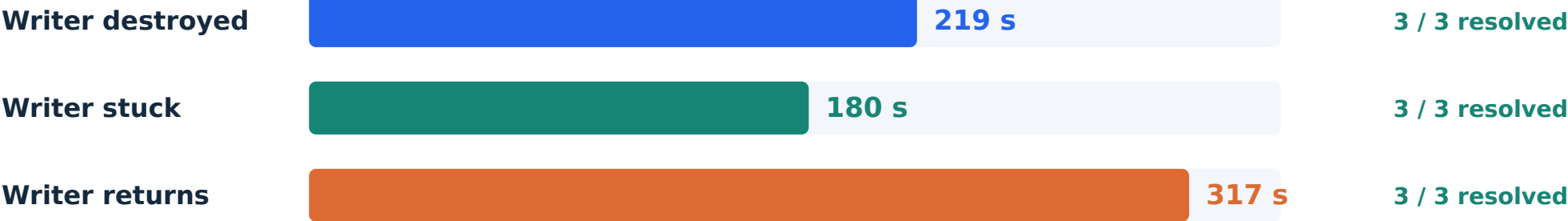
Assist by priority + update

Three Writer outcomes. Continued objectives.

08 / SIMULATION EVIDENCE

Three handovers. The objectives continued.

4 October 2026 · one recorded run per scenario · elapsed to objective completion



Fire + two victims · return journeys excluded from these timings · logs in docs/results/

3 / 3

Fire + victim memories resolved in each recorded scenario

149

Central automated tests passed 5 October 2026

2–3

Recorded RF hops to the gateway

Make failure visible. Preserve useful knowledge.

Writer loss

Independent beacon store.
Mission handover with available memories.

Relay loss

Alternative mesh routes.
One recorded relay removal recovered five dependent routes.

Link outage

Keep the latest state.
Forward on recovery and retry commands.

What still needs work

Physical sensing and radio measurements. Gateway redundancy. Explicit partial-success status and confirmed write-back. Return-clearance handling.

A completion banner alone is not proof: inspect target results, RESOLVED memories and return details

Keep the logic. Replace the simulated inputs.

09 / PHASE 2 PLAN

Replace simulated inputs with physical systems.

Planned milestones · two required robot bases · final deadline 1 December 2026

● 06-19 OCT

Firmware + RF

Frames / measured links

● 20-26 OCT

Robot bases

Odometry / LiDAR / SLAM

● 27 OCT-02 NOV

Dispenser + sensors

20 drops without a jam

● 03-09 NOV

Writer autonomy

Mock-tunnel exploration

● 10-16 NOV

Gateway + Executor

Brief over real radio

● 17-23 NOV

Failure trials

Loss / relay / outage

● 24 NOV-01 DEC

Final delivery

Manual / demo / pitch

REQUIRED PROTOTYPE

Writer + Executor

Real sensors, dispenser, battery beacons and gateway.

PLANNING BUDGET

≈ 4,650 DT

Two robots with 10% contingency; optional fire robot excluded.

The next robot starts with knowledge.

Explore → Preserve → Continue → Update

If the Writer is lost, the map is not.

[Read the six-page report ↗](#)

[Inspect recorded evidence ↗](#)

[Reproduce the simulation ↗](#)